

Recommended Assessment

Ultrasonic

Exploring Ultrasonic Sensing

1. How does the distance measurement on the scope change as you move the object closer to and further away from the device? What range of distances were accurately detected? How does this compare to the operating range stated on the datasheet? How consistent is the measured distance reading when you leave the object at a specific distance from the sensor?
2. How does the distance measurement change as you move the object at different speeds, or place it at different angles relative to the sensor? What if the object is not directly placed in front of the sensor?
3. How did the readings change when you decreased the ultrasonic sensor range? When would you want a smaller or larger detection range?

4. What was shown in the amplitude plot when you placed a water bottle 0.4 m from the sensor? Describe how different properties of objects placed in front of the sensor affected the shape of the amplitude plot. How did the color, shape, size, and material of the objects affect the amplitude of the measured signal? How does the signal appear for the same object at different distances? Include some screenshots of the amplitude scope, noting the object that was placed and its distance from the sensor.
5. How does the information from the datasheet explain observations from this lab? Discuss any factors that influenced the operating range.

Ultrasonic Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	